Zihan Qi

- **Mobile: (852) 61948792**
- **Email: normalman743@gmail.com**
- **GitHub: github.com/normalman743**
- **LinkedIn: zihan-qi-115b15301**

Education

The Chinese University of Hong Kong | 2022 - Present (Expected: July 2026)

BSc in Statistics, Minor in Computer Science | GPA: 3.8 / 4.0

Honors & Awards:

- Dept. of Statistics Scholarship： 2023/24 & 2024/25 & 2025/26
- Dean's List & Chung Chi College Head's List: 2024–2025; 2025–2026
- Chung Chi College Departmental Prize — Top academically performing final-year student; 2025/26

Relevant Coursework:

Data Structure(A), Computer Organization(A), Intro to Operating Systems(A), Web Applications (A)

Technical Skills

Programming: Python, R, C/C++, SQL, JavaScript, HTML/CSS

ML & DL Frameworks: TensorFlow, scikit-learn, PyTorch

Data Analysis & Visualization: pandas, NumPy, Excel, ggplot2, seaborn, Tableau

Databases: MySQL, MongoDB

Tools & Platforms: Git, Linux, Jupyter Notebook, VS Code, Terminal

LLM Api & RAG Tools: OpenAI (GPT), Anthropic Claude, Google Gemini; FAISS, Chroma

Research & Project Experience

Research Project– Large Models & Multimodal Alignment **CUHK · Sep 2024 – Present**

- Initiated this project under academic guidance from Prof. Xiangyu Yue (MMLab, CUHK), who also connected the project with the first author of OneLLM and several senior researchers for early-stage discussions on alignment methodology and experimental design.
- Conducted preliminary alignment experiments on OneLLM using a curated subset of the Flickr30k_images dataset to probe image–text matching behavior.
- Modified the OneLLM codebase to hook and exported. Early results show overall cosine similarity rising with depth, with discrimination peaking in mid layers (≈20) before flattening again and occasional deep-layer outliers.

Software Engineer Intern – RAG Backend & Admin Console Platform | Remote · Apr 2025 – Sep 2025

- Upgraded an autism-support Q&A platform to a Flask-based RAG backend (OpenAI embeddings, FAISS, MySQL), powering 600+ articles / 2,000+ segments / 10k+ bilingual question templates.
- Implemented an SSE streaming chat API with logging and API-key access control; designed multi-threaded FAISS initialization and hot updates (monthly scheduled + on-demand) to ingest new content and regenerate question–paragraph links.
- Built and shipped a full-stack internal admin console (content CRUD, dashboards, auto-Q generation, log analysis, sync/export) for non-technical staff; delivered most flows end-to-end.
- Ran a 2,400-call HK/US × EN/ZH benchmark: -32–34% avg latency, -58% worst-case, 99.9% success.

Others:

- SyncBridge (CSCI3100, 2025): Delivered full SRS/SDD + UML and a structured test plan/report for a requirements-negotiation web app; supported team integration and reviews.
- Full-Stack Cultural Event Platform (2023–2024): Built REST APIs (Node.js/Express, MongoDB) and a React/TypeScript frontend for auth, event CRUD, and geo-search over 100+ events.
- Auto Essay Grading (2023–2025): Prototyped a zero-shot LLM grading workflow across several hundred submissions; 93/104 students preferred "solution + AI feedback."
- Multi-Disease Prediction (2024): Processed a 4GB dataset and evaluated deep-learning baselines vs classical models across subgroups.